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The investigation of the influence of increasing mass on the drag coefficient of the ping pong ball at terminal velocity

Introduction

In the topic of sports, it is often mentioned how important is to improve performance and many sports could be related to Physics. In the case of table tennis, the speed of the ball can often play an important part in the result of the game. The speed of the ball and the way a player should interact with the ping-pong ball can be analyzed by a sub-field of Physics – Aerodynamics. As an aerodynamic object, a ping-pong ball should have a drag coefficient determining its performance (NASA, 2023) Depending on a ping-pong ball, they can have different masses, affecting the player's performance. To measure how the drag coefficient of a ping pong ball is related to mass, the following research question can arise: What is the effect of increasing mass (2.0, 4.0, 6.0, 8.0 g) on the drag coefficient at terminal velocity as measured by the tracking of the position of the ball during free fall?

Background theory

Drag force in free fall

When an object falls through the air or any fluid, the air or fluid exerts a force on the object opposite to the direction of the body's velocity relative to the fluid. The mentioned force is called **drag force** (NASA, 2018). The following equation can represent the drag force of an object falling through air or any fluid:

$$F_{drag} = \frac{1}{2} C_d A \rho V^2 \quad (1)$$

Where:

C_d – drag coefficient [dimensionless]

A – the frontal area [m^2]

ρ – density of fluid through which the object falls [kg m^{-3}]

V – velocity of the object [m s^{-1}]

Terminal velocity

At some point during the fall, the drag force will become equal to the weight, and the net force on the falling object will be zero (NASA, 2018).

$$W = mg \quad (2)$$

Where: m – mass of the object [kg]

g – gravitational field strength [m s^{-2}]

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The velocity at that point will be constant – the object will achieve a terminal velocity. The equation can be now written the following way, where the velocity became the terminal velocity.

$$W = \frac{1}{2} C_d A \rho V^2 \quad (3)$$

Where:

V – terminal velocity of the object [m s^{-1}]

Which, knowing that weight is equal to mass times gravitational field strength, can be further rearranged as:

$$C_d [-] = \frac{2gm}{V^2 A \rho} [-] \quad (4)$$

According to that equation, the hypothesis of the experiment is that the greater the mass of the ping-pong ball, the greater its drag coefficient. To be exact, the drag coefficient of the ping-pong ball should be proportional to its mass.

The measurement of mass will be accomplished by injecting a different mass of water inside the ping-pong ball using a syringe. The velocity of the ping-pong ball will be measured in computer software by tracking the position of the ball in each frame of the video of the ball's flight. Based on that, a graph of displacement against time will be produced, from which it will be possible to model its motion and find the terminal velocity. The terminal velocity will be used to calculate the drag coefficient for each mass.

Parameters

Independent Variable: The independent variable in the investigation will be the mass of the ping pong ball m which will be changed.

- The chosen method for changing the mass of the ball is adding respectively 0.0 g, 2.0 g, 4.0 g, 6.0 g, and 8.0 g of water inside the ball through the syringe.
- The masses of water added through the syringe are chosen to be relatively small - between 2.0 g to 8.0 g - as the ball itself is very light and otherwise could not achieve terminal velocity in the short 2-meter flight.

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- The water will be injected through the syringe as this way there is little risk that the syringe will alter the shape of the ball and large control over the exact added mass is possible.

Dependent Variable: The dependent variable in the investigation will be the drag coefficient C_d .

- The chosen method for measuring the drag coefficient C_d will be obtaining the terminal velocity of the ping pong ball and calculating the C_d from the equation (4).
- Measuring of the terminal velocity will be readily available via the use of the "Tracker" software (Douglas Brown, 2020).
- To make sure that only the terminal velocity is considered in the investigation, the ball will be dropped from 2.0 m, as it is the largest height from which the ball can be dropped in the laboratory. Only the last 0.1 m of flight will be analyzed in the "Tracker" software to make sure the velocity is as close to the terminal as possible.

Control Variables:

- **The size and shape of the ball**

  The area of the ball A facing the airflow must stay constant for the experiment to be valid and consistent with equation (4). The same ball will be used for all trials to avoid a random error.
- **The density of fluid through which the ball will be falling**

  The fluid through which the object is falling will be air. All measurements will be conducted at the same place at the same temperature to avoid any changes in density, so the measurements are consistent with equation (4).
- **How the ball is dropped**

  The ball will be dropped without introducing any force, to make sure it will be able to achieve free fall in the last 0.1 m of flight.
- **The height from which the ball is dropped**

  The ball will be dropped from the same constant height for all trials, to make sure it can achieve terminal velocity for all trials, as the ball can achieve higher speed while falling for a longer distance. The height will be 2.0 m for all trials, which will be measured with a meter ruler.

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- **The parallax error**

  The volume of water taken by the syringe will be measured at eye level to prevent taking the wrong amount of water by looking at the syringe from different viewing angles.
- **The syringe used**

  The same syringe will be used throughout all trials to prevent excessive holes or dents from forming on the surface of the ball.
- **Water on the ball**

  The water on the ball will be dried with a cloth between all trials to make sure not to introduce additional mass on the surface of the ball.
- **Sudden air movements**

  The experiment will be conducted in a lab where the doors and windows will be closed, preventing any sudden air flow movements from disrupting the ball's free fall.

Apparatus

- 1 ping-pong ball
- 1 2.5 g syringe (± 0.1 g)
- Electronic balance (± 0.01 g)
- Meter rule (± 0.01 m)
- Camera – iPhone 11
- Vernier calipers (± 0.0001 m)

Procedure

1. The mass of the ball was measured on an electronic balance and recorded in the experiment log.
2. The size of the ball was measured by using Vernier calipers and recorded in the experiment log.
3. The camera was set to record.
4. The ball was placed at the height of 2.0 m, positioning the last 0.1 m of the ball's expected flight trajectory in the view of the camera.
5. The ball was loosely dropped.
6. The ball hit the ground.
7. The recording on the camera was stopped and saved.

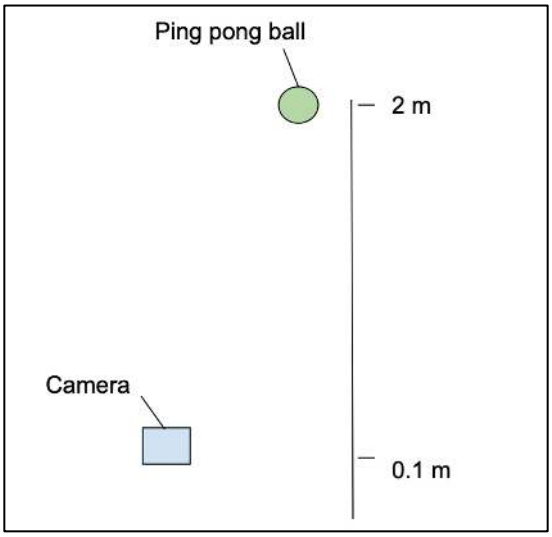


Figure 1. Experimental setup of the

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8. The video was edited to remove all parts when the ball was not in the view of the camera.
9. The video was uploaded into Tracker software.
10. The Tracker software produced a graph of the ball's changing position with time.
11. The slope of the produced graph was calculated.
12. The average slope during the last 0.1 m of fall was calculated and assumed to be the terminal velocity.
13. The steps 1-10 were repeated 5 times.
14. 2.0 g of water was measured and injected into the ball through the syringe.
15. The steps 3-14 were repeated 4 times.

Environmental Concerns

All the equipment used was reusable and no waste was created during the experiment.

Ethical Concerns

There were no ethical concerns detected in the investigation as there is no living organisms or moral issues involved in the experiment.

Safety Concerns

The syringe was handled with care as it was a very sharp needle. It was made sure that no water from the syringe or the cylinder met any of the electronic devices used in the course of the experiment.

Raw Data

Ping-pong ball data

	Mass Δ ± 0.01 [g]	**Radius Δ ± 0.001 [m]**
Ping-pong ball	2.74	0.021

Table 1. A summary of properties of the used ping-pong ball

The obtained values reflect a high degree of accuracy of the used equipment.

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Experiment raw data

Mass of water added Δ ± 0.01 [g]	Terminal velocity Trial 1 Δ ± 0.1 [m/s]	Terminal velocity Trial 2 Δ ± 0.1 [m/s]	Terminal velocity Trial 3 Δ ± 0.1 [m/s]	Terminal velocity Trial 4 Δ ± 0.1 [m/s]	Terminal velocity Trial 5 Δ ± 0.1 [m/s]	Mean velocity [m/s]
0.00	0.947	0.926	0.953	0.963	0.999	0.958 ± 0.037
2.00	1.067	1.079	1.014	0.996	1.056	1.043 ± 0.042
4.00	1.104	1.106	1.111	1.100	1.103	1.105 ± 0.006
6.00	1.120	1.123	1.135	1.129	1.138	1.129 ± 0.009
8.00	1.155	1.220	1.149	1.144	1.146	1.163 ± 0.038

Table 2. Raw data showing terminal velocity from each trial and the resulting mean velocity

The uncertainty of the balance was used instead of the uncertainty of the syringe with which the water was added.

To obtain the average terminal velocity the following formula was used, on the example of the 2.00 g case:

$$V_{avg} = \frac{V_1 + V_2 + V_3 + V_4 + V_5}{5} =$$

$$\frac{1.067 + 1.079 + 1.014 + 0.996 + 1.056}{5} = 1.043\ [m\ s^{-1}]$$

While calculating the uncertainty the formula below was used, on the example of the 2.00 g case:

$$\Delta V = \frac{V_{max} - V_{min}}{2} = \frac{1.067 - 0.996}{2} = 0.042\ [m\ s^{-1}]$$

All values in the table were formatted to 3 decimal places, except the values for the electronic balance as in this case only 2 decimal places were provided.

For the cases of m = 4.00 and 6.00 g, a very small uncertainty is present. This can suggest a random error present in other measurements.

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Processed Data

The density of air is quoted to 1.300 kg/m3.

Mass of the ball Δ ± 0.01 [g]	Mean velocity [m/s]	Drag coefficient [-]
2.74	0.958 ± 0.037	32.5 ± 5.7
4.74	1.043 ± 0.042	47.5 ± 8.4
6.74	1.105 ± 0.006	60.1 ± 6.5
8.74	1.129 ± 0.009	74.7 ± 8.4
10.74	1.163± 0.038	86.5 ± 13.8

Table 3. Data showing the calculated drag coefficient along the terminal velocity for each mass

The values of the drag coefficient were rounded to 1 decimal place. The drag coefficient was calculated using Equation 2. For the case of m = 2.74 g, the calculation appears as follows:

$$C_d = \frac{2gm}{V^2 A \rho} = \frac{2 \times 9.81 \times 2.74 \times 10^{-3}}{1.300 \times \pi \times 0.021^2 \times 0.958^2} \approx 32.5$$

The uncertainty for the calculated drag coefficient can be obtained using the following formula:

$$\frac{\Delta C_d}{C_d} = \frac{\Delta m}{m} + 2\left(\frac{\Delta r}{r}\right) + 2\left(\frac{\Delta V}{V}\right)$$

For the case of m = 2.74 g, it can be calculated exactly as follows:

$$\Delta C_d = 32.5 \times \left(\frac{0.01}{2.740} + 2\left(\frac{0.001}{0.021}\right) + 2\left(\frac{0.037}{0.958}\right)\right) = 5.7$$

In Table 3, it can be noticed that there is some form of positive correlation between the drag coefficient for the ping pong ball and the mass of the ball. Another important point is that many of the uncertainties in the drag coefficient are over 10% indicating that there is a consistent source of large error in the data.

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Graphical Analysis

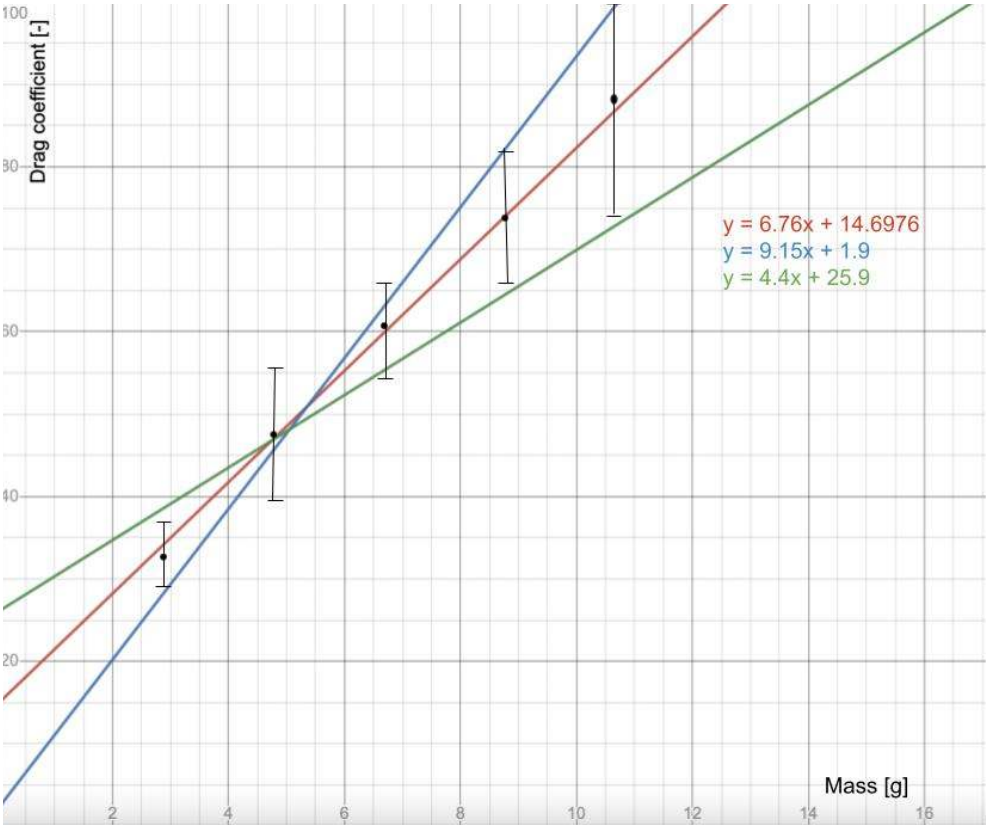


Figure 1. The obtained lines of best, minimum, and maximum fit for the data (Desmos).

The line of best fit accurately represents all the measured data points by visual inspection. The horizontal error bars were neglected due to small errors in mass measurement. The calculated R^2 coefficient for the best-fit line is 0.9986, which indicates that there is a very strong positive correlation between the drag coefficient and mass of the ping-pong ball in each trial. The mass of 10.74 g presents the largest error bar in the drag coefficient, which indicates the presence of a large random error in this case.

The uncertainty of the gradient can be calculated using the following formula:

$$\Delta k = \frac{k_{max} - k_{min}}{2} = \frac{9.15 - 4.4}{2} = 2.375$$

$$k = 6.76 \pm 2.375$$

The uncertainty of the y-intercept can be calculated in the following way:

$$\Delta_{intercept} = \frac{25.9 - 1.9}{2} = 12$$

$$y_{intercept} = 14.7 \pm 12\ m^2\ s^{-2}$$

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The percentage uncertainty of the gradient can be calculated using the formula below:

$$\Delta_{\%\ uncertainty\ of\ gradient} = \frac{\Delta k}{k} \times 100\% = \frac{2.375}{6.76} \times 100\% = 35\%$$

It is important to mention that the obtained linear fit of the model can be used in a limited number of cases as both the mass along drag coefficient must always be positive and non-zero according to laws of Physics and commonly agreed conventions.

Conclusion

While answering the research question **"What is the effect of increasing mass (2.0, 4.0, 6.0, 8.0 g) on the drag coefficient at terminal velocity as measured by the tracking of the position of the ball during free fall?"** it was found that there is a strong positive linear correlation between increasing mass of the ball and its drag coefficient.

Even though a strong positive linear correlation was found, it should not necessarily be the case as the drag coefficient should not change as rapidly, considering the small increments of mass that were added in each trial.

Another point of reflection for this study could be the large values themselves, obtained for the drag coefficient. In the literature the values of C_d usually obtained for all sorts of objects are supposed to be between 0.1 and 2 (NASA, 2023). In the study the values of drag coefficient oscillate between 32 – 86. This suggests a large source of error present at some point in the investigation. A possible source of this error could be the fact that the dropped ping-pong ball did not manage to reach a terminal velocity during the 2.0 m fall and the taken terminal velocity is too small. This would explain the large overshoot in the drag coefficient value.

The large uncertainties in the drag coefficient could be additionally attributed to the large uncertainties in mean terminal velocity, which can be explained by the imprecise method of measurement of the terminal velocity. The values of terminal velocities were relatively different from each other in all the 5 trials, contributing to the large error in mean velocity and subsequently the drag coefficient.

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Evaluation

Strengths

The strengths of the investigation could be a large number of trials performed (5) leading to minimizing the random human errors during the measurement. Another strength of the experiment could be using the same syringe which effectively increases the mass of the ball while not largely affecting the area size of the ball. What's more, the large experimental precision was obtained by measuring the mass using accurate equipment and actively minimizing the parallax error while adding mass to the ping-pong ball.

Limitations and improvements

Weakness	Impact	Improvements
Low resolution of the video recording of the ball's flight. (systematic error)	This could produce inaccurate displacement graphs in the software.	Using a camera with higher resolution, as the program could more accurately locate the ball, as well as track it, which would produce accurate velocity graphs.
The low height from which the ball was dropped. (systematic error)	This could lead to the ball not achieving terminal velocity during its flight.	The ball could be dropped from a higher altitude so there is certainty that it has achieved terminal velocity, and the proposed formulas are appropriate to be used.
Weakness	**Impact**	**Improvements**
Adjusting the 1 metre reference distance for measuring the displacement in the Tracker 5.1.5. (systematic error)	The program directly measured the displacement and velocity in relation to the reference points, so the distances might have been inaccurately measured.	Possibly using a specialized camera stand to ensure that image in the camera is not tilted and that the distance is accurately measured in the video, producing accurate terminal velocity values.

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Wrong orientation of the camera. (systematic error)	The wrong orientation of the camera might have distorted the image and therefore caused wrong measurements of displacement.	Every camera distorts the image to some extent, so using a photocell detecting movement could be the most relevant way to solve that issue and get accurate displacement results.
Disruptions of air flow in the room where the experiment was conducted. (random error)	It could have influenced the path chosen by the ball and therefore disrupt the calculations of velocity and displacement performed by the Tracker software.	Conducting the experiment in a room without any windows, as even closed windows might produce unnecessary airflow where airflow couldn't be disrupted, so by improving this, all the velocities measured could be more accurate.

Extensions

To further enhance the investigation and add an additional research perspective, different types of balls could be used and the influence of the type of ball on the drag coefficient of the ball could be investigated. The relationship could be mathematically modeled and then conclusions could be drawn about which type of ball is most efficient aerodynamically, apart from a ping pong ball.

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Video editing software “Tracker 5.1.5” 13 May 2020 by © Douglas Brown

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